

The Value Added by Accountants

Teaching note

Dechow (1994), *Journal of Accounting and Economics*

ACCTG 597A · Fall 2026 · Sam Bonsall · Penn State

The net effect of accruals is an empirical question

“Information about enterprise earnings and its components measured by accrual accounting generally provides a better indication of enterprise performance than does information about current cash receipts and payments.”

FASB, SFAC No. 1 ¶44, as quoted in Dechow (1994, p. 4)

“Many financial analysts regard operating cash flow as a better gauge of corporate financial performance than net income . . . ”

Chemical Week (1991), as quoted in Dechow (1994, p. 5)

Her reframing (p. 5): “it is an empirical question as to whether the net effect of accruals is to improve or reduce the ability of earnings to measure firm performance.”

Stock returns benchmark the comparison

Information asymmetries create demand for a periodic summary measure; realized cash flows are the primitive candidate, but timing and matching problems make them noisy, and accruals are hypothesized to mitigate both. The question is relative: contracts pick one measure.

With markets assumed efficient (p. 12), the decision rule follows:

“The performance measure (either realized cash flows or earnings) that has a higher association (R^2) with stock returns is then interpreted as more effectively summarizing firm performance.”

Dechow (1994, p. 13)

Controlling for timing and matching problems “would defeat the purpose of the test” (p. 14).

A one-parameter collection model carries the intuition

$$C_t = (1 - \varphi)S_t + \varphi S_{t-1}$$

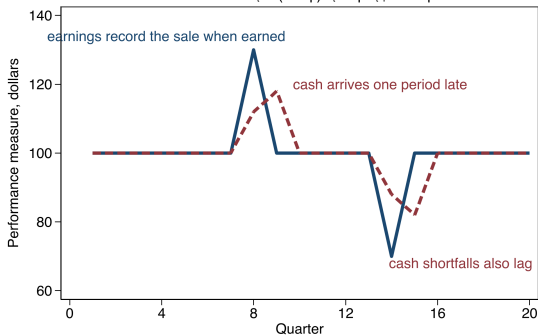
$$S_t - C_t = \varphi \Delta S_t$$

The gap between earnings and cash receipts, the accrual, grows in the credit proportion φ (the operating cycle) and in the volatility of activity $|\Delta S_t|$ (eqs. 1-2, pp. 9-10).

The accrual records exactly the timing difference; over long intervals the temporary components cancel.

Cash receipts lag the performance they measure

Collection model $C_t = (1 - \varphi)S_t + \varphi S_{t-1}$ with $\varphi = 0.6$



Four hypotheses map timing problems into testable orderings

	Prediction	Tested in
H1	Earnings associate more strongly with returns than cash flows over short intervals	Tables 3-4
H2	The cash flow association improves relative to earnings as the interval lengthens	Table 3
H3	The earnings advantage grows with the absolute magnitude of aggregate accruals	Table 5
H4	The cash flow association declines with operating-cycle length; earnings less sensitive	Table 6

Two further component predictions: working-capital accruals, not long-term operating accruals, do the mitigating work (Table 7), and special items reduce earnings' short-window association (Table 8).

The manipulation alternative signs H2 the other way: if accrual discretion degrades earnings, earnings should improve relative to cash flows as the window lengthens.

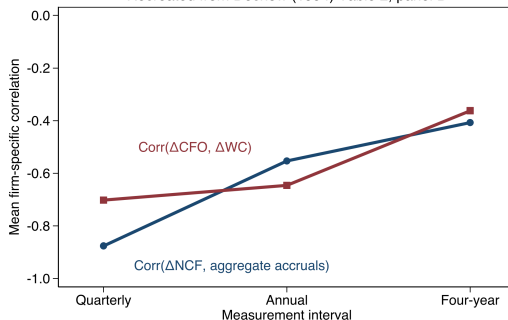
Accruals covary negatively with cash flow changes

Cash flow changes mean-revert sharply (autocorrelation -0.52 ; earnings -0.18): they carry large temporary components that accruals absorb.

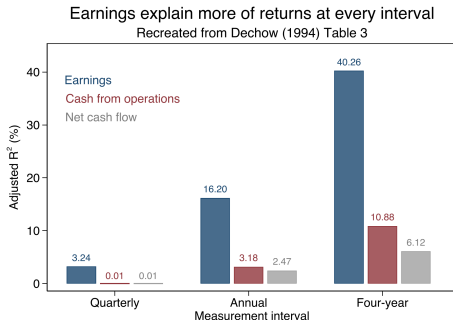
$\text{Corr}(\Delta\text{NCF}, \text{accruals})$ is -0.88 quarterly, weakening to -0.41 over four years as matching problems abate.

Smoothing would produce the same correlation (p. 19); Table 2 alone cannot adjudicate.

Accrual/cash-flow correlations weaken with the interval
Recreated from Dechow (1994) Table 2, panel B



Cash flows explain almost nothing over short intervals



The ordering earnings > CFO > net cash flow holds at every interval (H1); the ratios R_{CFO}^2/R_E^2 rise from 0.003 to 0.20 to 0.27 (H2), a partial convergence.

Comparing the fit of nonnested models requires the Vuong (1989) test

“However, simply comparing R^2 ’s does not provide statistically reliable evidence that earnings is superior to cash flows.”

Dechow (1994, p. 23)

The structure that makes the comparison valid: both regressions explain the *same dependent variable* (stock returns) on the *same sample*, so their likelihoods are commensurable. Vuong’s test selects the model closer to the truth “without presuming under the null that either model is ‘true’ ” (p. 23).

When both measures carry incremental power, the Davidson-MacKinnon J-test “lacks power and cannot distinguish between the competing hypotheses” (pp. 23-24).

Every cash flow measure is rejected in favor of earnings

Interval	E vs. CFO	E vs. NCF	CFO vs. NCF
Quarterly	7.60	7.62	0.06
Annual	24.16	23.27	2.27
Four-year	17.04	18.03	3.67

Vuong Z -statistics, Table 4; a positive Z rejects the second measure in favor of the first. Earnings comparisons all $p < 0.001$; CFO vs. NCF is insignificant quarterly ($p = 0.954$).

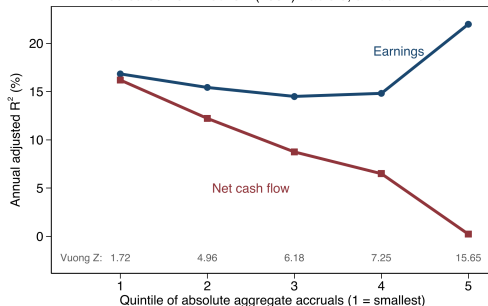
Appendix 2 supplies the template later papers copied, and the borrowed technology spread: Biddle-Seow-Siegel (1995), Dhaliwal-Subramanyam-Trezevant (1999), and Barth-Cram-Nelson (2001) adopt the test; Collins-Maydew-Weiss (1997) do not require it: their decomposition is nested.

The earnings advantage grows with accrual magnitude

In the smallest-accrual quintile the measures are indistinguishable (Vuong $Z = 1.72$); in the largest, net cash flow's R^2 collapses to 0.24% while earnings holds 21.98% ($Z = 15.65$).

The operating-cycle evidence points the same way (Table 6): across 58 industries, the CFO R^2 correlates -0.483 ($p = 0.001$) with cycle length; the earnings correlation is -0.083 ($p = 0.538$).

Cash flows lose explanatory power as accruals grow
Recreated from Dechow (1994) Table 5, annual interval



Several central comparisons rest on ordered magnitudes, not tests

Claim	Evidence	Absent test
H2: cash flows converge toward earnings	R^2 ratios rise $0.003 \rightarrow 0.20 \rightarrow 0.27$ (Table 3)	Change in the ratio across intervals
Matching problems abate with horizon	$\text{Corr}(\Delta\text{NCF}, \text{AA}): -0.88 \rightarrow -0.55 \rightarrow -0.41$ (Table 2)	Difference of correlations
Earnings insensitive to operating cycle	-0.483 (CFO) vs. -0.083 (E) (Table 6)	The two correlations against each other
WC accruals outrank long-term accruals	R^2 paths across two separate sorts (Table 7)	Interaction test of the two gradients
Robustness footnotes	results asserted unchanged (fns. 13, 14, 17)	No test statistics reported

The Vuong test operates within a measurement interval; no analogous test bridges intervals. The paper's own p. 23 standard applies with equal force to the comparisons above.

The second-order-manipulation conclusion is a synthesis, not a test

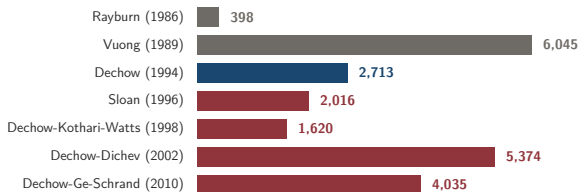
“This suggests that management manipulation of accruals is of second-order importance and the first-order effect of the accrual process is to produce a summary measure that more closely reflects firm performance.”

Dechow (1994, p. 28)

The sentence rests on the joint pattern of Tables 2-5, not on a statistic that isolates the manipulation channel; Dechow acknowledges that the design “does not provide tests to separately determine the extent of” manipulation versus matching (pp. 13-14, 19).

The distinction matters for what came next: Dechow-Dichev (2002) and the discretionary-accruals literature begin exactly where this sentence stops.

One design seeded the accrual literature



Citations (OpenAlex, August 2026); gray = antecedent and borrowed test, navy = the focal paper, maroon = the literature it seeded; Sloan (1996) understated

AAA Seminal Contributions to Accounting Literature Award, 2019. The verdict is retested on prediction: Nallareddy-Sethuraman-Venkatachalam (2020) find cash flows overtaking earnings in predicting future cash flows; Ball-Nikolaev (2022) attribute the reversal to cash flow measurement and restore the earnings ranking.

Summary

- Dechow documents that the accrual system induces a strong negative correlation between accruals and cash flow changes, and that earnings dominate realized cash flows in explaining returns over short intervals, consistent with accruals mitigating timing and matching problems rather than primarily enabling manipulation.
- Her methodological discipline set a standard: competing nonnested regressions with the same dependent variable are compared with the Vuong (1989) likelihood-ratio test, a borrowed technology the literature adopted from this paper's appendix.
- The paper also shows that a seminal contribution can mix formally tested claims with ordered-magnitude readings; the cross-interval convergence, the correlation contrasts, and the second-order-manipulation conclusion rest on comparisons no statistic adjudicates.

Dechow (1994) · Vuong (1989) · Sloan (1996) ·
github.com/sbb151/acctg597a-research-design